

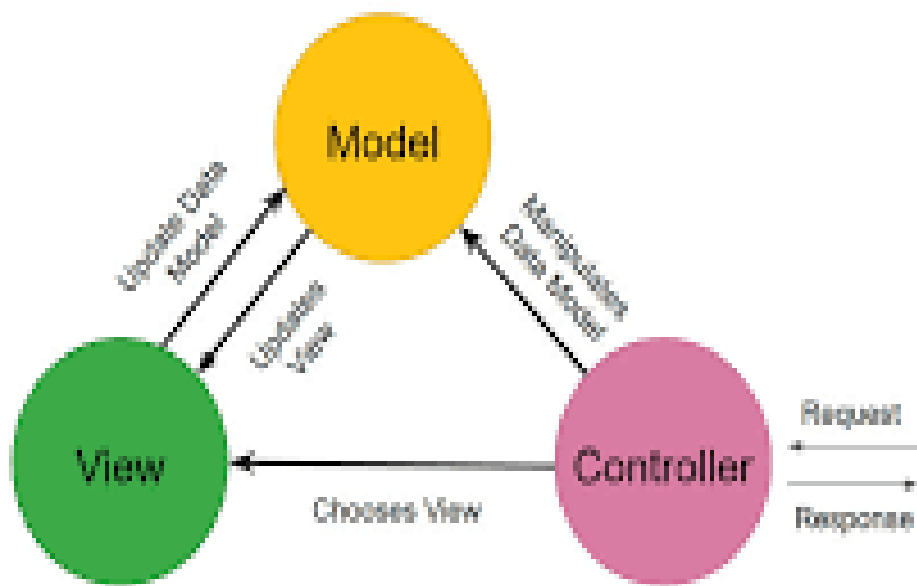
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ASSIGNMENT – 1

Design Pattern Explanation - Prepare a one-page summary explaining the MVC (Model-View-Controller) design pattern and its two variants. Use diagrams to illustrate their structures and briefly discuss when each variant might be more appropriate to use than the others.



1 What is MVC?

MVC is a software architectural pattern used for developing user interfaces that divides an application into three interconnected components:

Model: Manages the data, logic, and rules of the application.

View: Displays the data of the user, it's the user interface.

Controller: Handles user input, updates the model and view.

2 Two Variants of MVC:

Traditional MVC (as used in desktop apps):

- User interacts with the view.
- View sends input to the controller.
- Controller updates the Model.
- Model changes reflect in the view.

MVC in web Applications:

- User interacts with web pages (View) through browser.
- Controller often resides in the server, it processes the request

- Model processes business logic and database operations.
- View is rendered server-side or dynamically updated via frontend JS framework.

3 Use Case Scenarios:

- Traditional MVC: Desktop applications like Java Swing, NET Windows Forms.
- Web MVC: Frameworks like Django, Ruby on Rails, ASP.NET MVC.